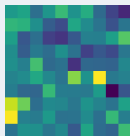


Input



SERS map

DQ24:TBZ12:THI6 μM

NNLS hit gate

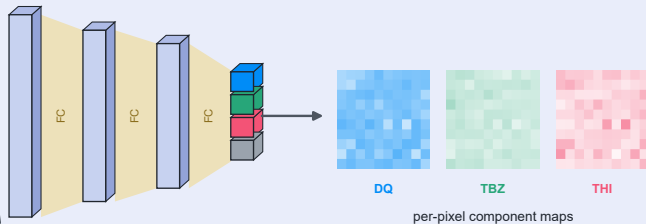


hit-pixel spectra

$\log(1+x) \cdot 1,290 \text{ ch}$

500–2500 cm^{-1}

Composition module — shared per-pixel MLP



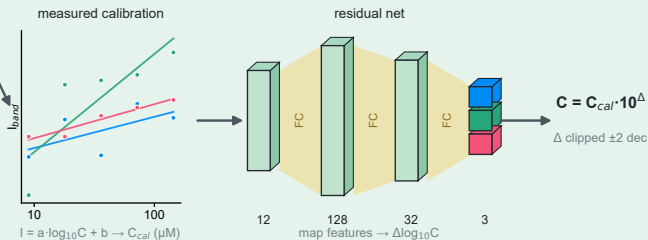
map composition

mean pool over hit pixels
BLK removed · renormalised

49% 24% 27%

probability-weighted band signal I_{eq} (1570 / 1270 / 1367 cm^{-1})

Quantification module — calibration-anchored residual



Output

concentration (μM)

per component · validated window · OOD flags

known total (optional): $C_i = p_i \times C_{total}$

declared metadata — constrained